

HYDABURG, AK, USA

WMO: 703884

Lat: 55.206N

Lon: 132.828W

Elev: 0

StdP: 14.70

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 25376

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	18.6	22.6	1.4	5.9	24.4	8.7	8.6	26.5	39.3	46.1	34.6	45.0	4.1	20	0.659

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.2	70.3	58.9	66.4	56.5	63.9	55.4	60.0	67.2	58.4	64.2	57.2	62.3	9.2	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
56.7	68.5	62.3	55.3	65.0	60.6	54.5	63.2	59.7	26.3	67.8	25.2	64.0	24.5	62.1	65.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 31.4	26.1	21.7	DB	13.0	73.0	4.9	5.5	9.5	77.0	6.7	80.2	4.0	83.3	0.5	87.3	(4)
(5)			WB	12.9	61.3	5.4	2.2	9.1	62.9	5.9	64.1	2.9	65.4	-1.0	67.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	46.1	37.2	37.9	38.8	42.8	47.7	53.2	56.8	57.5	53.3	47.8	41.1	38.1	(6)	
	DBStd	8.78	7.30	5.22	4.36	3.80	4.07	3.72	3.22	2.91	4.22	5.32	5.23	6.10	(7)	
	HDD50	2149	396	338	347	219	95	6	0	0	14	99	267	368	(8)	
	HDD65	6912	861	758	812	665	536	355	256	234	351	535	716	833	(9)	
	CDD50	715	0	0	0	4	23	102	210	232	112	31	1	0	(10)	
	CDD65	6	0	0	0	0	0	1	2	1	0	2	0	0	(11)	
	CDH74	31	0	0	0	0	0	14	12	5	0	0	0	0	(12)	
	CDH80	2	0	0	0	0	0	2	0	0	0	0	0	0	(13)	
(14) Wind	WSAvg	7.5	8.3	7.6	8.0	7.5	7.0	6.7	6.3	6.1	7.4	8.3	8.3	8.7	(14)	
(15) Precipitation	PrecAvg	104.8	11.5	8.6	8.4	6.4	4.6	3.8	4.8	6.7	10.7	13.4	12.9	13.0	(15)	
	PrecMax	142.7	21.4	17.8	15.8	9.8	11.5	7.0	8.9	14.6	19.3	20.9	23.1	28.2	(16)	
	PrecMin	75.3	3.5	3.1	3.0	0.6	0.3	1.4	0.4	1.7	3.9	5.9	5.9	5.9	(17)	
	PrecStd	17.0	3.6	3.9	3.3	2.3	2.6	1.4	2.0	3.0	3.7	3.4	4.1	5.4	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.4	53.6	53.6	63.9	67.6	74.8	75.2	73.4	67.5	69.7	54.7	52.0	(19)	
		MCWB	42.6	42.6	44.4	48.2	52.6	58.2	62.4	60.7	56.3	57.0	46.1	45.8	(20)	
	2%	DB	48.3	48.2	48.2	55.5	62.1	66.7	70.1	69.7	63.5	65.7	51.8	48.4	(21)	
		MCWB	44.3	43.4	42.6	44.9	50.4	55.8	59.8	59.5	54.9	53.3	47.0	44.7	(22)	
	5%	DB	46.7	46.1	46.3	52.0	57.3	63.3	65.6	65.7	61.1	55.5	49.5	47.6	(23)	
		MCWB	43.9	42.2	41.7	44.2	48.8	54.1	57.3	58.0	54.4	51.0	46.1	44.7	(24)	
	10%	DB	45.8	45.2	45.4	48.3	54.7	59.4	62.9	63.2	58.8	54.1	47.9	45.9	(25)	
		MCWB	43.0	41.4	41.3	43.0	48.0	52.6	55.7	56.3	53.8	50.3	45.1	43.1	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	46.5	46.3	46.3	50.7	54.7	59.3	63.2	62.2	59.2	56.5	49.7	47.4	(27)	
		MCDB	47.5	48.8	49.2	58.5	62.8	69.2	74.0	71.3	63.4	66.2	51.5	48.7	(28)	
	2%	WB	46.0	44.8	44.7	47.1	51.6	56.7	60.2	60.1	57.1	53.8	47.5	46.3	(29)	
		MCDB	47.0	46.7	47.1	53.5	58.4	65.6	68.1	66.7	60.9	59.7	49.2	47.4	(30)	
	5%	WB	44.9	43.0	43.1	45.7	50.3	54.8	58.2	58.7	55.8	52.4	46.2	45.2	(31)	
		MCDB	46.3	46.1	46.2	49.6	56.2	61.3	63.9	64.1	59.5	57.0	48.1	46.5	(32)	
	10%	WB	43.3	41.5	41.5	44.2	49.0	53.3	56.7	57.6	54.7	50.6	45.2	43.3	(33)	
		MCDB	45.8	45.3	45.1	48.1	54.2	58.3	61.6	62.2	58.0	54.0	47.3	45.8	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	7.2	8.3	8.7	10.4	10.2	9.6	8.9	9.2	9.1	8.0	7.4	7.1	(35)	
		MCDBR	7.4	8.8	10.6	17.6	17.2	17.7	17.1	16.6	13.6	8.8	7.8	6.5	(36)	
	5% WB	MCWBR	6.0	6.3	7.3	9.6	8.9	8.8	8.9	8.6	7.8	6.6	6.4	5.1	(37)	
		MCWBR	6.6	7.3	8.5	12.5	13.8	14.7	13.9	12.8	10.0	6.6	7.4	6.3	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.306	0.316	0.341	0.360	0.366	0.364	0.374	0.361	0.361	0.338	0.313	0.307	(40)	
		taud	2.361	2.475	2.420	2.394	2.426	2.470	2.475	2.536	2.489	2.462	2.329	2.242	(41)	
	Ebn at Noon	Ebn at Noon	200	246	264	274	279	280	274	272	255	233	191	163	(42)	
		Edh at Noon	17	21	28	33	34	33	32	28	26	21	17	15	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	148	393	718	1174	1468	1542	1370	1195	793	432	199	116	(44)	
	RadStd	RadStd	20	71	98	72	142	132	164	119	100	55	25	20	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (16 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+108	N/A

Nomenclature: See separate page